

Topic 7.2

7.2: Abstract Data Types (ADTs)

There are times when we write a class with the expectation that no object of that class type will ever be created. The sole purpose for such a class is to serve as the common base for many derived classes.

We can prevent objects of such a class from being created by specifying that one of the class's virtual member functions is *pure*:

```
class some_ADT
{
    ...
    // Pure virtual function.
    virtual int some_member_func() = 0;
    ...
}
```

Such a class is often referred to as an abstract data type (ADT). The following program fragment illustrates the idea of ADTs.

```
#define    PI 3.14159

class Shape    // An ADT.
{
    public:
        // Pure virtual function
        virtual double circumference() = 0;
        // Pure virtual function
        virtual double area() = 0;
        // Pure virtual function
        virtual void print() = 0;
}

class Circle : public Shape
{
    public:
        Circle( double r, double xc, double yc )
            { radius = r; x = xc; y = yc };
        double circumference()
            { return 2 * PI * radius; };
        double area() { return PI * radius * radius; };
        void print() { cout << "Circle of radius
                        << radius <<
                        " at (" << x << ", " << y << ")";};

    protected:
        double radius; // Radius of circle.
        double x;      // X-coordinate of circle's center
        double y;      // Y-coordinate of circle's center
}
```

```
class Rectangle : public Shape
{
    public:
    Rectangle( double lp, double wp, double xp,
    double yp )
    { l=lp; w=wp; x=xp; y=yp; };
    double circumference() { return 2 * (l + w); }
    double area() { return l * w; }
    void print() { };

    protected:
    double l;    // Length (longer dimension).
    double w;    // Width (shorter dimension).
    double x,y;  // X-Y Coordinates of
    // lower left corner.
}

void main()
{
    //Shape s;           // THIS WON'T WORK!
    Circle   c(2.5,3,2); // Declare a circle.
    Rectangle r(7,9,5,3); // Declare a rectangle.

    cout << "The area of ";
    c.print();
    cout << " is " << c.area() << endl;
}
```

Topic 7.3 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-3>)